UNIVERSITY OF TORONTO
Faculty of Arts & Sciences
April 2019 Examinations

MAT136H1

Duration: 3 hours

Aid Allowed: Scientific Calculator

Multiple Choice Question Sheet

- One calculator that meets the posted guidelines may be used. No other aids are permitted.
- This is the multiple choice Question Booklet for the test. It consists of 9 pages, including this cover sheet.
- This section contains Questions 1–20.
- Each multiple choice question is worth 2 marks. No partial credit will be awarded.
- **Do not write answers on this paper**. Any question sheets collected will be destroyed immediately after the exam.
- **RECORD** your answers, using the bubbles on the final page of the examination paper. You need to write dark enough for a computer to be able to read your answers.
- You are also required to complete Questions 21–27 on the other portion of the exam.
- Select only one answer for each question.
- Questions appear on both sides of the page.

Faculty of Arts & Sciences Guidelines:

- Do not begin writing the actual exam until the announcements have ended and the Exam Facilitator has started the exam.
- Turn off and place all cell phones, smart watches, electronic devices, and unauthorized study materials in your bag under your desk. If it is left in your pocket, it may be an academic offence.
- When you are done your exam, raise your hand for someone to come and collect your exam. Do not collect your bag and jacket before your exam is handed in.
- If you are feeling ill and unable to finish your exam, please bring it to the attention of an Exam Facilitator so it can be recorded before leaving the exam hall.

Formula Sheet

- Geometric Formulas
  - Surface area of sphere: $A = 4\pi r^2$
  - Volume of sphere: $V = \frac{4}{3}\pi r^3$
  - Surface area of cone (including base): $A = \pi r(r + \sqrt{h^2 + r^2})$
  - Volume of cone: $V = \pi r^2 \frac{h}{3}$
  - Surface area of cylinder (including top and base): $A = 2\pi rh + 2\pi r^2$
  - Volume of cylinder: $V = \pi r^2 h$
- Taylor series centred at $x = 0$ (note that the interval of convergence is *not* given):
  - e^x: $\sum_{n=0}^{\infty} \frac{x^n}{n!}$
  - $\sin(x)$: $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!}$
  - $\cos(x)$: $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$
  - $\frac{1}{1-x}$: $\sum_{n=0}^{\infty} x^n$
  - $\ln(1+x)$: $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$
  - binomial series $(1+x)^p$: $1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \cdots$

1. (2 points) True or False: If a power series of the form $\sum_{k=0}^{\infty} a_k x^k$ converges at $x = 10$ then it must converge at $x = -9$. Note that the a_k are constants.

   A. True

   B. False

2. (2 points) Which integral is represented by the following limit?

$$\lim_{b \to 1^-} \left(\lim_{n \to \infty} \sum_{i=0}^{n-1} \frac{1}{\sqrt{1 - x_i}} \Delta x \right)$$

   where $\Delta x = \frac{b}{n}$ and $x_i = i \Delta x$.

   A. $\displaystyle\int_1^{\infty} \frac{1}{\sqrt{1-x}} dx$

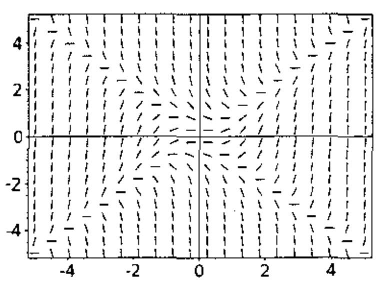
   B. $\displaystyle\int_1^{\infty} \frac{x}{\sqrt{1-x}} dx$

   C. $\displaystyle\int_0^{1} \frac{1}{\sqrt{1-x}} dx$

   D. $\displaystyle\int_0^{1} \frac{x}{\sqrt{1-x}} dx$

   E. none of the above

3. (2 points) Which of the following differential equations corresponds to the slope field shown below?



   A. $\dfrac{dy}{dx} = (x - y)^2$

   B. $\dfrac{dy}{dx} = (x + y)^2$

   C. $\dfrac{dy}{dx} = x^2 - y^2$

   D. $\dfrac{dy}{dx} = x^2 + y^2$

In their paper *The intra-push velocity profile of the over-ground racing wheelchair sprint start*, Moss et al. recorded the velocity and momentum of an international male wheelchair athlete during a push. The velocity of the racer was increasing during the first 0.6 seconds of the push. The following table shows the athlete's velocity $v(t)$, measured in m/s, t seconds after the push began.

t (s)	0	0.2	0.4	0.6
$v(t)$ (m/s)	0	0.10	0.65	1.40

The function $v(t)$ described is used in questions 4 and 5.

4. (2 points) How often would the athlete's velocity need to be measured in order to ensure that a left-hand sum approximating the total distance traveled between $t = 0$ and $t = 0.6$ was within 0.7 m of the actual value? Choose the greatest value that satisfies this criteria.

   A. every 0.1 s

   B. every 0.25 s

   C. every 0.5 s

   D. every second

   E. none of the above

5. (2 points) Let $M(t)$ be the momentum of the athlete measured in N·s (Newton-seconds) t seconds after he began the same recorded push. In the integral

$$\int_0^2 M(v^{-1}(x))dx$$

what are the units of the 2?

   A. seconds

   B. metres

   C. Newtons

   D. metres per second

   E. Newton-seconds

6. (2 points) Which one of the following sequences diverges as n approaches ∞?

   A. $\left\{1 + \frac{1}{n}\right\}$

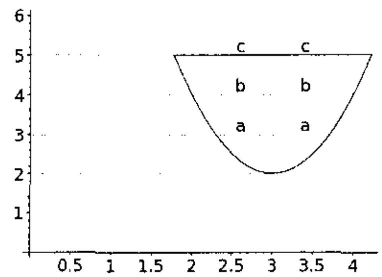
   B. $\left\{\frac{1-n}{n}\right\}$

   C. $\left\{\frac{\cos(n\pi)}{n}\right\}$

   D. $\{n\cos(n\pi)\}$

   E. none of the above sequences diverge

7. (2 points) We can generate a solid S by rotating the shape shown below about the y-axis.



Which of the following is a Riemann sum that approximates the volume of this solid S?

A. $\pi((3+a)^2 + (3+b)^2 + (3+c)^2)$

B. $\pi((3-a)^2 + (3-b)^2 + (3-c)^2)$

C. $\pi((3+a)^2 + (3-a)^2 + (3+b)^2 + (3-b)^2 + (3+c)^2 + (3-c)^2)$

D. $\pi((3+a)^2 - (3-a)^2 + (3+b)^2 - (3-b)^2 + (3+c)^2 - (3-c)^2)$

E. None of the above

Determine whether the series in questions 8 and 9 converge or diverge.

8. (2 points) $\displaystyle\sum_{n=1}^{\infty} \frac{6n}{\sqrt{n^2+n+5}}$

A. this series converges

B. this series diverges

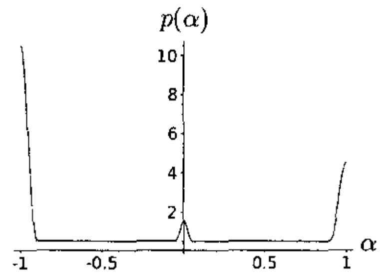
9. (2 points) $\displaystyle\sum_{n=5}^{\infty} \frac{(\ln n)^4}{n}$

A. this series converges

B. this series diverges

In the book *Complex spreading phenomena in social systems*, the authors analyze the number of times Italian Facebook users *like* different posts. Suppose that user number u has performed x likes on scientific posts and y likes on conspiracy posts. Then they define the *polarization* $\alpha(u) = \dfrac{y-x}{x+y}$ as a measure of how much the user likes scientific posts versus conspiracy posts. A user u is said to be polarized towards science if $\alpha(u) \leq -0.95$ and polarized towards conspiracy if $\alpha(u) \geq 0.95$.

The probability density function (PDF) $p(\alpha)$ of Italian Facebook users' polarization is shown below. In this question 'user' will refer to Italian Facebook users included in the study.



The probability density function described is used in questions 10 through 13.

10. (2 points) According to the graph of the PDF, which of the following is greatest?

    A. the probability that a user has polarization between -0.05 and 0

    B. the probability that a user has polarization between 0 and 0.05

    C. the probability that a user is polarized towards science

    D. the probability that a user is polarized towards conspiracy

11. (2 points) Assuming that $p(\alpha)$ is indeed a PDF, which of the following are equal to $\int_{-\infty}^{\infty} p(\alpha)d\alpha$?

    A. 0

    B. 1

    C. $\int_{-1}^{0} p(\alpha)d\alpha$

    D. $\int_{0}^{1} p(\alpha)d\alpha$

12. (2 points) What can you say about the value of the integral $\int_{-1}^{1} \alpha p(\alpha)d\alpha$?

    A. it is negative

    B. it is equal to 0

    C. it is positive

    D. we cannot conclude any of the above from the information given

Note that Question 13 refers to the function $p(\alpha)$ on the previous page.

13. (2 points) Let $P(\alpha)$ be the cumulative distribution function related to $p(\alpha)$. We say that a user u is *inclined towards science* if $\alpha(u)$ is negative. Which of the following represent the probability that a user is inclined towards science?

    A. $P(-1)$

    B. $P(0)$

    C. $P(1)$

    D. $\int_{-1}^{0} P(\alpha)d\alpha$

    E. $\int_{0}^{1} P(\alpha)d\alpha$

14. (2 points) Which of the following functions satisfies the differential equation $\frac{d^2y}{dx^2} = -16y$?

    A. $y = e^{x/4}$

    B. $y = e^{-x/4}$

    C. $y = \sin(4x) + \cos(4x)$

    D. $y = 4\cos x - 4\sin x$

15. (2 points) True or False:

$$\frac{d}{dt}\int_1^{t^2} \frac{\sin(x)(x+2)^5}{2^x}dx = \frac{\sin(t^2)(t^2+2)^5}{2^{t^2}}.$$

    A. True

    B. False

16. (2 points) The Taylor series centred at $x = 0$ of the function $\cos x$ is

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}.$$

Suppose that you now want to prove that the Taylor series converges for every value of x. You already know that the Taylor series centred at $x = 0$ of $\sin x$ converges for every value of x. Two of your friends propose the following arguments:

Alex: Apply the Ratio Test to the $\cos x$ Taylor series

Arna: Differentiate the $\sin x$ Taylor term-by-term, and use the fact that this preserves the radius of convergence

Which of these arguments is complete?

A. Alex's argument is complete, but Arna's is not

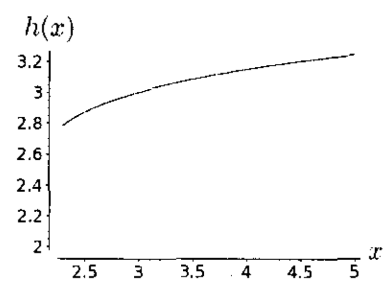
B. Arna's argument is complete, but Alex's is not

C. Both Alex's and Arna's arguments are complete alone

D. Alex's and Arna's arguments can be used together to provide a complete argument, but neither one is complete on its own

E. None of the above

17. (2 points) The function $h(x)$ is a function whose graph is shown below. Suppose that a Taylor polynomial centred at $x = 3$ is used to approximate $h(3.1)$. Four classmates who have been given a table of values for h and various derivatives produce four different sums to approximate $h(3.1)$ using this Taylor polynomial; all but one can be eliminated as the correct answer with just the graph. Which of the following sums could be correct?



A. $3 + 0.71(0.1) + \dfrac{0.71(0.1)^2}{2!} + \dfrac{0.71(0.1)^3}{3!}$

B. $3 + 0.71(3.1) + \dfrac{0.71(3.1)^2}{2!} + \dfrac{0.71(3.1)^3}{3!}$

C. $3 + 0.71(0.1) - \dfrac{0.71(0.1)^2}{2!} - \dfrac{0.71(0.1)^3}{3!}$

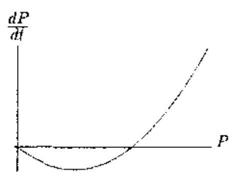
D. $3 + 0.71(3.1) - \dfrac{0.71(3.1)^2}{2!} - \dfrac{0.71(3.1)^3}{3!}$

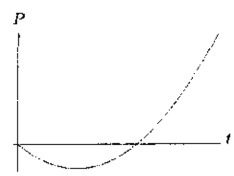
A population of animals relies solely on chance encounters between males and females when mating. Suppose that the population is given by $P(t)$. The rate of growth will be proportional to the product of the number of males, $P/2$, and the number of females, $P/2$ and the *death rate* may be assumed to be a constant δ. Under these assumptions, the differential equation modelling our population is

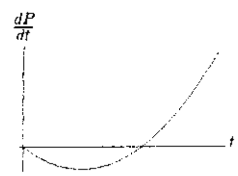
$$\frac{dP}{dt} = kP^2 - \delta P = kP(P - \alpha)$$

where k, δ, and α are positive constants such that $\alpha = \delta/k$. Questions 18 through 20 refer to this population model.

18. (2 points) Which of the following graphs corresponds to the population model described?

(i) 

(ii) 

(iii) 

A. (i) only

B. (ii) only

C. (iii) only

D. (i), (ii), and (iii)

E. None of the above.

19. (2 points) Suppose that when $t = 0$, $P < \alpha$. What happens to the population in the long-term?

A. $\lim_{t\to\infty} P(t) = 0$

B. $\lim_{t\to\infty} P(t) = \alpha$

C. $\lim_{t\to\infty} P(t) = \delta$

D. $\lim_{t\to\infty} P(t) = \infty$

E. None of the above

20. (2 points) Suppose that when $t = 0$, $P > \alpha$. What happens to the population in the long-term?

A. $\lim_{t\to\infty} P(t) = 0$

B. $\lim_{t\to\infty} P(t) = \alpha$

C. $\lim_{t\to\infty} P(t) = \delta$

D. $\lim_{t\to\infty} P(t) = \infty$

E. None of the above

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MAT136H1

Duration: 3 hours
Aid Allowed: Scientific Calculator

LAST (Family) NAME: ____________________

FIRST (Given) NAME: ____________________

STUDENT NUMBER: ____________ **Section:** ________

EMAIL: ____________________@mail.utoronto.ca

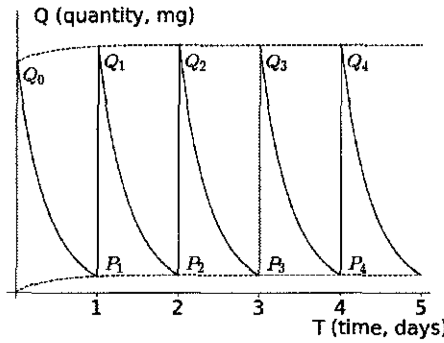
This exam contains 10 pages (including this cover page and the multiple choice question sheet) and a SEPARATE multiple choice question booklet. Check to see if any pages are missing. **100 points are possible.**

- One calculator that meets the posted guidelines may be used. No other aids are permitted.
- **Questions appear on both sides of the page.**
- **Indicate your answers for Questions 1–20 on the multiple choice question sheet, by shading the bubble corresponding to your answer.** Leave the remaining bubbles blank. The Question Sheet will not be collected or graded.
- Do not write in the page margins. Make sure that your writing is dark enough to be readable.
- **Unsupported answers to Questions 21–27 will not receive full credit.** A correct answer without explanation will receive no credit unless otherwise noted; an incorrect answer supported by substantially correct calculations and explanations may receive partial credit.
  - **Include units** in your answers where appropriate. Round to 3 decimal places.
  - **Organize your work** in a reasonably neat and coherent way.
  - You must use the methods learned in this course to solve all of the problems.

21. (6 points) Determine ~~the endpoints of~~ the interval of convergence of

$$\sum_{n=0}^{\infty} \frac{n(x+2)^n}{2^{n+1}}.$$

22. (8 points) The drug Atenolol is prescribed to Chris to lower blood pressure. Suppose that Chris takes one 50 mg dose each morning at exactly 8 a.m. The figure below shows the quantity of the drug in the blood as a function of time. Suppose that the first dose is taken at $t = 0$, where t is measured in hours, and that the half-life of Atenolol is 6.3 hours. (A half-life is the time it takes for the amount of drug to decay by one-half.)



Write your final answer to each part on the lines provided.

(a) (2 points) What percentage of Atenolol in the blood at the start of a 24-hour period is still there at the end?

percentage remaining = ____________

(b) (4 points) As shown in the figure, $Q_0, Q_1, Q_2, Q_3, \ldots$ are quantities of drug present in Chris's blood right after they take their 8 a.m. dose each morning. Find a general expression for Q_n in closed form. 'In closed form' means that your answer should only depend on n (not Q_n), and should not include any summation signs.

$Q_n =$ ____________

(c) (2 points) As shown in the figure, $P_1, P_2, P_3, \ldots$ are quantities of drug present in Chris's blood immediately before taking the 8 a.m. dose each morning. Find a general expression for P_n in closed form.

$P_n =$ ____________

23. (10 points) The paper *Cigarette Smoke in Closed Spaces*, published by Ulrich Hoegg in 1972, quantified the impact of people smoking cigarettes on non-smokers in the same closed public spaces. One of the substances that is released into the air when people smoke is carbon monoxide. It can cause headaches and dizziness in people when it makes up more than 0.02% of the air.

Researchers tested the concentration of substances in a 25 m^3 test chamber with air-circulating fans. In one experiment, air containing 5% carbon monoxide (this means that 5% of the **volume** of the incoming air is carbon monoxide) was introduced at a rate of 0.003 m^3/min. Assume that the carbon monoxide mixes with the rest of the air immediately, and the mixture leaves the room at the same rate as it entered.

(a) (2 points) The experimenters decide to keep going with the experiment until the concentration of carbon monoxide is greater than 0.02%. Will such a level ever be reached? (Circle one)

Yes No

Give a one-sentence explanation for your answer that does not involve a computation.

(b) (3 points) At what rate is the volume of carbon monoxide, Q, entering the room at t minutes?

(c) (3 points) At what rate is the volume of carbon monoxide, Q, leaving the room at t minutes?

(d) (2 points) Use your answers to parts (b) and (c) to write a differential equation for the rate of change of the volume of carbon monoxide with respect to time t. Your answer may only contain the variables t and Q.

$$\frac{dQ}{dt} = \underline{\hspace{8cm}}$$

24. (7 points) Let $f(t) = \ln(t^2)$.

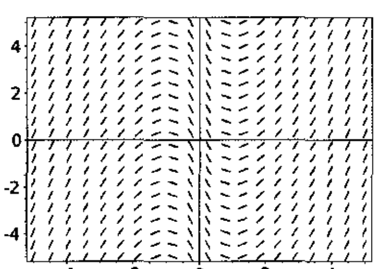
(a) (4 points) Consider the differential equation

$$\frac{dy}{dx} = f(x).$$

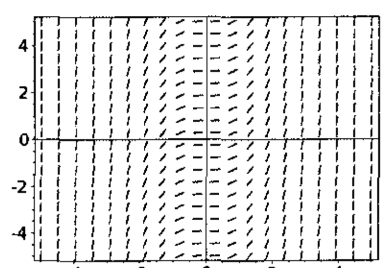
Use Euler's method with $\Delta x = 0.1$ to fill-in the rest of this table of values for the function y.

x	3	3.1	3.2	3.3
y	0			

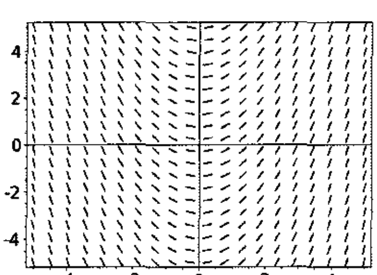
(b) (3 points) Which of the following slope fields corresponds to the differential equation $\frac{dy}{dx} = f(x)$? Circle the letter corresponding to the correct slope field.



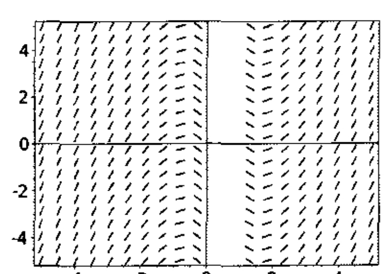
A.



C.



B.



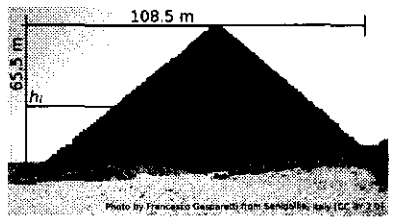
D.

25. (8 points) A 12-foot-tall artificial Christmas tree has circular cross sections at every height. It is decorated so that the tree's weight density varies as you move up the tree. The following table shows the factory specifications for the radius and weight density (in pounds per cubic foot) of the tree at different heights, measured in feet above the ground.

Height (feet)	0	3	6	9	12
Radius (feet)	8	5	4	2	0
Weight Density (pounds / ft^3)	5.2	4.8	3.3	2.1	N/A

Use the data in the table to write a left-hand Riemann sum to estimate the total weight of the tree, and write your final estimate in the box below. Be sure to explain where all of the elements in your solution come from.

26. (13 points) The Pyramid of Menkaure is one of the three main pyramids of Giza. It has a height of 65.5 m and a base of 108.5 m. In this problem, you will show how to compute the volume of material required to make the pyramid, assuming that it is a perfect square-based pyramid.



The initial setup is the partition of the height interval $[0, 65.5]$ into n intervals of equal size Δh with endpoints h_i.

(a) (4 points) Write a formula for $\ell(h_i)$, the length of one side of a square cross section of the container h_i metres above the base (bottom of the pyramid).

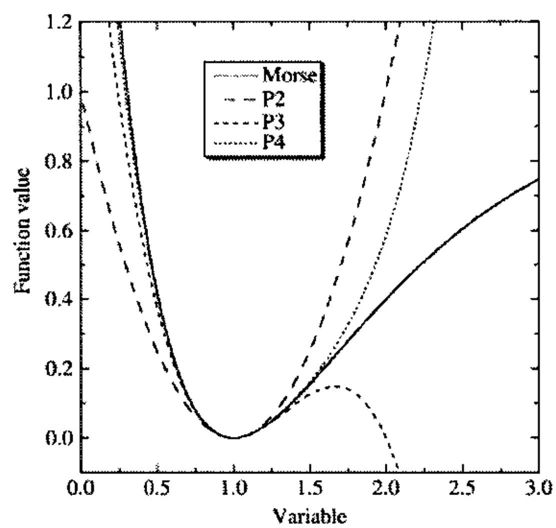
(b) (3 points) Write a sum that gives the approximate volume of a slice with thickness Δh located h_i metres above the base.

(c) (2 points) Use your answer from (b) to write an expression that represents the approximate volume of the pyramid.

(d) (4 points) Write an integral that computes the exact volume of the pyramid. Be sure to carefully explain how it follows from your previous work. You do not need to evaluate your integral.

27. (8 points) Suppose that in CHM220 you are studying the Morse potential. The Morse potential describes the energy of the system as a function of the distance between atoms. Chemists use this model to understand and predict chemical reactions, as it provides information about the vibrational energies required to pull the atoms far enough apart that the chemical bond will break. Your lab manual says

> We will consider the Morse potential in reduced units:
> $$y_{\text{Morse}} = (1 - e^{(x-1)})^2.$$
> The figure shows the second-, third-, and fourth-order Taylor approximations to the Morse potential.



Question 27 is continued on the next page.

This is a continuation of Question 27.

Some students in the class have only completed MAT135, and tell your professor that they cannot make sense of this passage. Your professor emails you:

> I noticed on your transcript that you passed MAT136. Can you write a short post on the class discussion board that explains this passage to your classmates who took MAT135 but not MAT136? Your explanation should answer the questions: what relationship do the curves in the diagram (Taylor polynomials) have to y_{Morse}? When would someone use them?
>
> If you write an explanation satisfying the following criteria, I'll award you a bonus grade on your final exam. Your explanation should:
>
> - clearly answer the two questions above
> - be mathematically correct and address both questions completely
> - take into account your classmates' background, and relate what they do not know with what they know
> - use complete sentences
> - short enough to fit into the space provided (while being easily readable)
>
> Thank you!

Write an explanation for your professor in the box below.

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1	A	B	C	D	E
2	A	B	C	D	E
3	A	B	C	D	E
4	A	B	C	D	E
5	A	B	C	D	E
6	A	B	C	D	E
7	A	B	C	D	E
8	A	B	C	D	E
9	A	B	C	D	E
10	A	B	C	D	E
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30	A	B	C	D	E
31	A	B	C	D	E
32	A	B	C	D	E
33	A	B	C	D	E
34	A	B	C	D	E
35	A	B	C	D	E
36	A	B	C	D	E
37	A	B	C	D	E
38	A	B	C	D	E
39	A	B	C	D	E
40	A	B	C	D	E